Sr.No. 8520

Bharatiya Vidya Bhavan's

# SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

## MID SEMESTER EXAMINATION

| | | |
|---|---|---|
| AY: 2025-26 | Session: Odd MSE October 2025 | Date: 08-10-2025 |
| Sem: 1 | Slot: 10:00am - 11:00am | C. Code: CS413 |
| Exam Type: Written | Block:401 Course Section:1 | Seat No: 2022600051 |

L_29_508_986_1_1505_38364

Name of the candidate. Pratiksha B. Salukte

Candidate's UID number: 2 0 2 2 6 0 0 0 5 1

Examination class room number: 4 0 1

Invigilator's signature with date: 8/10/25

**(To be filled in by the candidate)**

EXAMINATION: MSE BTech
(For example FY MCA/FY BTECH)

SEMESTER : 7

I / II / III / IV / V / VI / VII / VIII

PROGRAM: BE-AIML

DAY AND DATE: Wednesday 8/10/25

COURSE NAME: DL

**(To be filled in by the examiner)**

| Question numbers | 1 | 2 | 3 | 4 | 5 | 6 | Total | Signature of the examiner |
|---|---|---|---|---|---|---|---|---|
| Marks awarded | | | | | | | | |
| Marks after revaluation | | | | | | | | |

8250

## INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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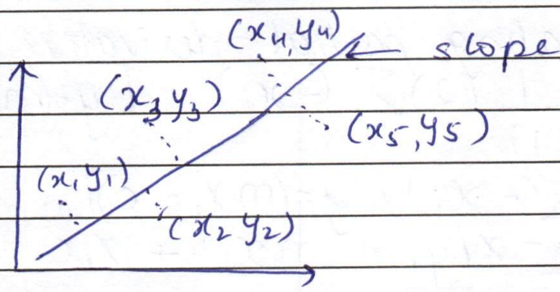
1 a)




There are 5 samples in the graph
A line is drawn to separate into different classes.

We know,

slope

$$y = mx_i + c \quad \therefore y = \sum_{i=1}^{n=5} mx_i + c$$

~~Least~~ squared error

$$\therefore y = \sum_{i=1}^{n=5} (mx_i + c)^2$$

Now calculate average/mean squared error

$$y = \frac{1}{n} \sum_{i=1}^{n=5} [mx_i + c]^2$$

Now using partial derivative, w.r.t. m

$$\frac{\partial y}{\partial m} = \frac{1}{n}(2) \quad [mx_i + c]$$

For error,

$$E = [y_i - \hat{y}_i]$$

Substitute the slope

$$E = [y_i - (mx_i + c)]$$

Calculating squared errors,

$$E = [y_i - (mx_i + c)]^2$$

Calculating average squared error

$$E = \frac{1}{n} \sum_{n=1}^{n=5} [y_i - (mx_i + c)]^2$$

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Calculating partial derivative wrt m.

$$\frac{\partial E}{\partial m} = \frac{1}{n}(2)\sum_{n=1}^{n}(-x_i)\,[y_i-(mx_i+c)]$$

$$= 0 = \sum_{n}(-x_i)[y_i-(mx_i+c)]$$

$$0 = \sum_{n=1}^{n}(-x_iy_i + mx_i^2 + x_ic_i)$$

$$\boxed{\therefore \sum_{n=1}^{n} mx_i^2 + \cancel{x_i}\sum_{n=1}^{n} x_ic = \cancel{x}\sum_{n=1}^{n} x_iy_i} \quad (1)$$

Calculating the partial derivative wrt. c

$$\frac{\partial E}{\partial c} = \frac{1}{n}(2)\sum_{n=1}^{n} \cancel{\alpha}(-)[y-(mx_i+c)]$$

$$= 0 = \sum_{n=1}^{n} -[y_i-(mx_i+c)]$$

$$0 = \sum_{n=1}^{n} -y_i + mx_i + c_i$$

$$\boxed{\therefore \sum_{i=1}^{n} mx_i + \sum_{i=1}^{n} c_i = \sum_{n=1}^{n} y_i} \quad -(2)$$

From eqn (1) and (2)

~~$y_{new} = y_{old} - \alpha\, \partial E$~~

$$\boxed{m = m - \frac{\partial E}{\partial m}}$$

$$\boxed{c = c - \frac{\partial E}{\partial c}}$$

$$\eta < \frac{2}{\lambda_{max}}$$

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1b) Overfitting happens when the training data is too complex due to which the model learns noise as well
In this the variance is very high.
It works well on training data but not on the test data.

As for underfitting, the data is too simple. ~~But~~ It doesn't learn important features ~~Du~~ due to which both variance and bias is very high.

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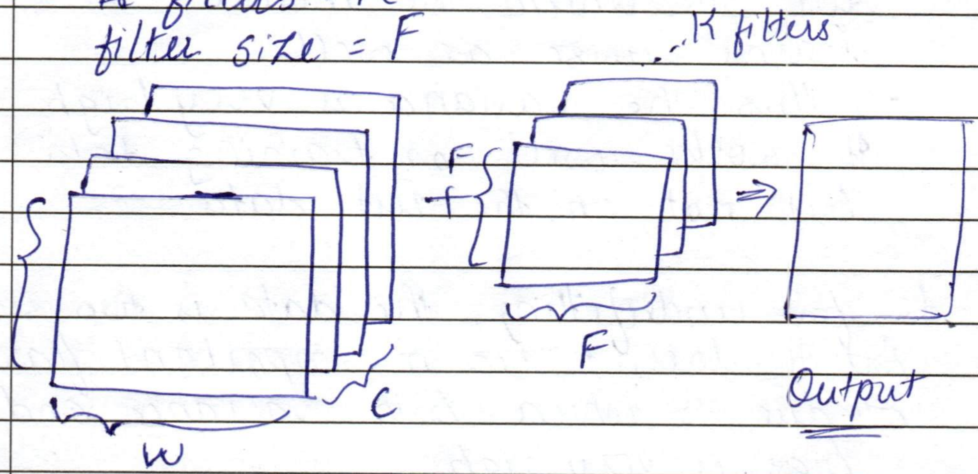
| | 2a) | For standard 2D convolutions |

input size : $H \times W \times C$
~~K~~ filters $= K$
filter size $= F$




To derive the output size we use.

$$O_{size} = \left[\frac{I - K + 2P}{S}\right] + 1$$

For calculating the ~~height~~ $O_H$ (Height) and since there is no mention of padding and stride.

The output =
$$O_H = [H - K] + 1$$

For width similarly
$$O_W = [W - K] + 1$$

Let no. of convolution layers of $(H \times M \times C)$ be $N$.

There are K filters.

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The computational complexity would be

$O(N \times K)$

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2b) Undercomplete encoder: is a dimensionality reduction algo which compresses the images to smaller units.

Here in the ~~la~~ hidden state the number of neurons are smaller

$$\dim(\text{input}) > \dim(\text{hidden})$$

Eg:-

$$100 \xrightarrow[\text{encode}]{} 20 \xrightarrow[\text{decode}]{} 100$$

It compresses the image due to which it learns important features only.

But the drawback is that compressing the images also causes it to loss ~~ea~~ important features.

It won't be able to learn certain important patterns.

input state

$$I = W_d x + b$$

hidden state

$$Y\ \cancel{I_h} = W_h x + b$$

Calculating compression (loss function)

$$\text{Output} = \| y - \hat{y} \|^2$$

target

$$= \| y - (W_h x + b) \|^2$$

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For overcomplete encoder, it has more neurons in hidden state compared to in the input state
$dim(input) < dim(hidden)$

Eg: $100 \xrightarrow[\text{encode}]{} 200 \xrightarrow[\text{decode}]{} 100$

In overcomplete encoder, the model learns more noise. While extracting the feature it also learns alot of noise since the number of neurons is high. Due to this it causes overfitting.
To avoid overfitting, regularization techniques are used.

input state:
$I = W_d x + b$

hidden state:

$Y_h = W_h x + b$

loss function
Output $= \|y - \hat{y}\|^2$
$= (y - \hat{y}) - \lambda \Omega$

strength

regularization technique

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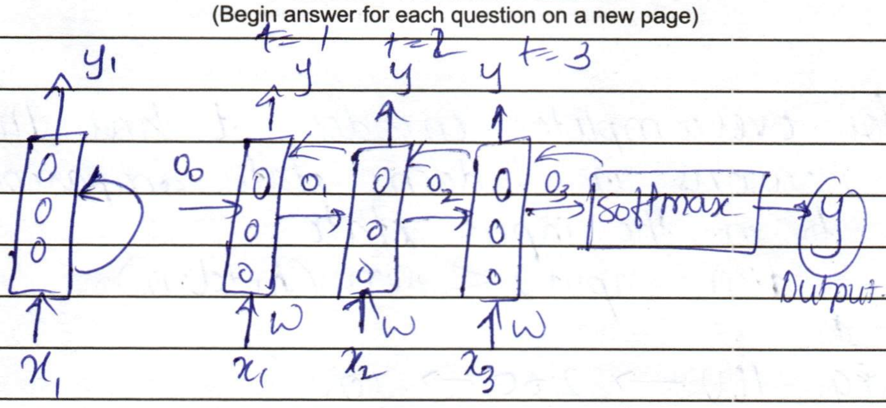
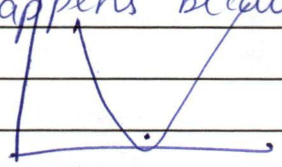
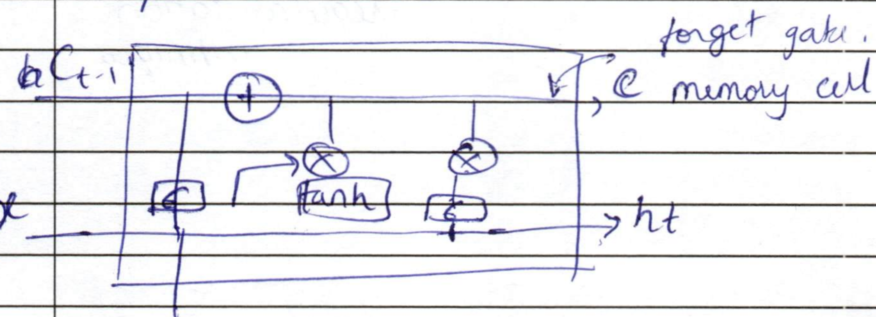
3a)




In RNN when back propagation takes place,
the weights are being updated
When the weights are getting updated, the feature extracted in the started loses its importance and the value become low leading it to not converge
This causing vanishing gradient problem
This happens because of sigmoid/tan h function.



In LSTM, due to the use of Forget Gate, it can keep track of all the important features
and thus vanishing gradient and exploding gradient does not take place.



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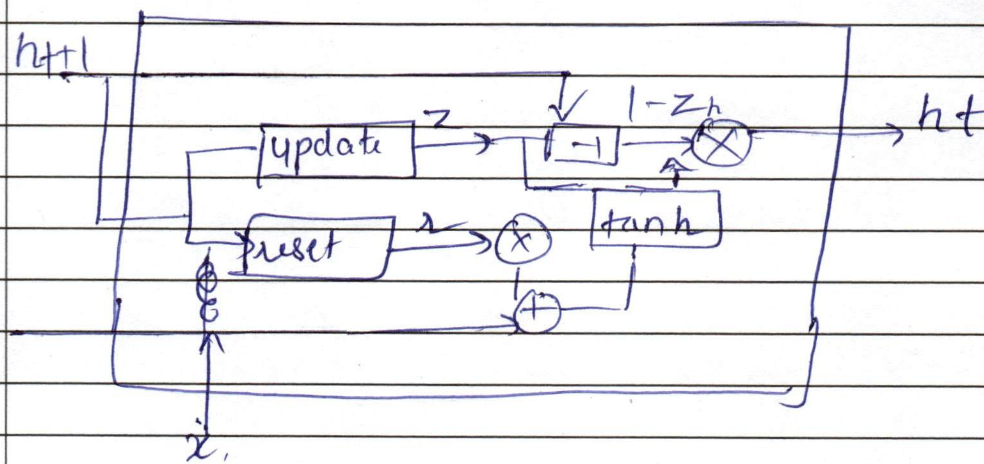
3b)




In GRU, there are only two gates and LSTM there is only ~~one~~ 3 gates

GRU keeps track of all the features while LSTM loses track of it after some time

Update gate

$z_h = \sigma(W_z x_i + U_z h_{t+1} + b)$

Reset gate

$r_h = \sigma(W_r x_i + U_r h_{t+1} + b)$

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